

## ESPERANCE AP, Australia

WMO: 956380

Lat: 33.6825S

Lon: 121.8275E

Elev: 143

StdP: 99.62

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 5.5        | 6.3 | -0.5                          | 3.7 | 18.3 | 1.3 | 4.2 | 16.3 | 13.8                  | 13.9 | 12.6 | 13.1 | 4.9                   | 0    | 0.592 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 10.8                   | 35.3            | 19.2 | 32.5 | 18.8 | 29.9 | 18.3 | 21.4                | 29.6 | 20.4 | 28.0 | 19.6 | 26.4 | 6.3                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.1                            | 14.2 | 23.1 | 18.2 | 13.4 | 21.9 | 17.5 | 12.7 | 21.2 | 63.0          | 29.8 | 59.3 | 28.0 | 56.4 | 26.3 | 27.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |      |     |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|------|-----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |      |     |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |     |
| (4) 12.1          | 10.7 | 9.7 | DB                         | 3.7 | 41.3               | 0.6 | 2.5                                                | 3.3 | 43.1       | 3.0 | 44.6       | 2.6 | 45.9       | 2.2 | 47.7 | (4) |
| (5)               |      |     | WB                         | 2.9 | 23.7               | 0.7 | 1.3                                                | 2.4 | 24.7       | 2.0 | 25.4       | 1.6 | 26.2       | 1.1 | 27.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 16.3   | 20.7  | 20.9  | 19.6  | 17.7  | 15.1  | 12.7  | 11.8  | 12.3  | 13.4  | 15.2  | 17.4  | 19.0  | (6)  |  |
|                                                                        | DBStd       | 4.32   | 3.41  | 3.18  | 3.27  | 3.03  | 2.54  | 1.87  | 1.82  | 2.33  | 2.64  | 3.22  | 3.48  | 3.38  | (7)  |  |
|                                                                        | HDD10.0     | 10     | 0     | 0     | 0     | 0     | 0     | 1     | 4     | 3     | 2     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD18.3     | 1100   | 9     | 7     | 21    | 47    | 107   | 168   | 203   | 188   | 151   | 110   | 61    | 28    | (9)  |  |
|                                                                        | CDD10.0     | 2307   | 331   | 306   | 299   | 232   | 157   | 83    | 59    | 74    | 103   | 161   | 221   | 281   | (10) |  |
|                                                                        | CDD18.3     | 356    | 82    | 80    | 61    | 29    | 6     | 0     | 0     | 1     | 2     | 12    | 32    | 51    | (11) |  |
|                                                                        | CDH23.3     | 3254   | 776   | 658   | 496   | 207   | 41    | 1     | 0     | 9     | 31    | 150   | 354   | 532   | (12) |  |
| (13)                                                                   | CDH26.7     | 1398   | 376   | 289   | 209   | 61    | 7     | 0     | 0     | 1     | 7     | 51    | 148   | 250   | (13) |  |
| (14)                                                                   | Wind        | WSAvg  | 5.3   | 5.6   | 5.5   | 5.1   | 4.7   | 4.9   | 5.3   | 5.5   | 5.4   | 5.2   | 5.5   | 5.6   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 549    | 23    | 28    | 34    | 38    | 69    | 68    | 81    | 72    | 56    | 46    | 38    | 18    | (15) |  |
|                                                                        | PrecMax     | 730    | 209   | 97    | 174   | 92    | 146   | 188   | 142   | 126   | 140   | 84    | 106   | 59    | (16) |  |
|                                                                        | PrecMin     | 324    | 0     | 0     | 0     | 4     | 13    | 28    | 18    | 20    | 0     | 13    | 3     | 0     | (17) |  |
|                                                                        | PrecStd     | 89     | 38    | 25    | 36    | 24    | 34    | 34    | 29    | 29    | 30    | 20    | 24    | 16    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 39.5  | 37.7  | 35.9  | 31.8  | 27.4  | 21.8  | 20.9  | 24.5  | 27.1  | 32.0  | 35.4  | 37.8  | (19) |  |
|                                                                        |             | MCWB   | 20.4  | 20.3  | 18.7  | 17.7  | 15.9  | 13.0  | 12.9  | 15.0  | 16.1  | 16.6  | 18.0  | 18.4  | (20) |  |
|                                                                        | 2%          | DB     | 35.1  | 33.3  | 32.0  | 28.2  | 23.8  | 19.7  | 18.7  | 20.7  | 23.1  | 27.3  | 31.0  | 32.8  | (21) |  |
|                                                                        |             | MCWB   | 19.7  | 19.9  | 18.5  | 17.1  | 14.9  | 13.0  | 12.6  | 13.6  | 14.6  | 16.1  | 17.6  | 18.4  | (22) |  |
|                                                                        | 5%          | DB     | 30.8  | 30.2  | 28.5  | 25.5  | 21.7  | 18.2  | 17.1  | 18.6  | 20.7  | 24.1  | 27.5  | 29.2  | (23) |  |
|                                                                        |             | MCWB   | 19.3  | 19.3  | 18.0  | 16.5  | 14.6  | 12.6  | 12.2  | 13.0  | 14.1  | 15.4  | 16.7  | 18.1  | (24) |  |
|                                                                        | 10%         | DB     | 27.4  | 27.2  | 25.5  | 23.1  | 19.7  | 16.8  | 15.7  | 16.8  | 18.5  | 21.3  | 24.2  | 25.7  | (25) |  |
|                                                                        |             | MCWB   | 18.6  | 18.9  | 17.7  | 16.1  | 14.0  | 12.1  | 11.7  | 12.3  | 13.2  | 14.7  | 16.1  | 17.3  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 22.9  | 23.4  | 21.4  | 20.3  | 18.6  | 15.9  | 15.2  | 16.2  | 17.2  | 18.7  | 20.3  | 21.3  | (27) |  |
|                                                                        |             | MCDB   | 32.2  | 30.7  | 28.6  | 26.1  | 23.4  | 19.2  | 17.5  | 21.4  | 23.8  | 26.3  | 30.0  | 29.3  | (28) |  |
|                                                                        | 2%          | WB     | 21.3  | 21.6  | 20.0  | 18.6  | 16.6  | 14.5  | 13.8  | 14.8  | 15.7  | 17.1  | 18.7  | 19.7  | (29) |  |
|                                                                        |             | MCDB   | 30.5  | 29.5  | 26.9  | 24.1  | 20.4  | 17.3  | 16.7  | 18.6  | 20.7  | 24.4  | 26.7  | 28.8  | (30) |  |
|                                                                        | 5%          | WB     | 20.1  | 20.4  | 19.2  | 17.7  | 15.6  | 13.6  | 12.9  | 13.7  | 14.6  | 16.1  | 17.6  | 18.7  | (31) |  |
|                                                                        |             | MCDB   | 28.1  | 27.1  | 24.8  | 22.7  | 19.4  | 16.4  | 15.8  | 17.3  | 19.0  | 22.2  | 25.0  | 26.9  | (32) |  |
|                                                                        | 10%         | WB     | 19.1  | 19.5  | 18.3  | 16.9  | 14.8  | 12.8  | 12.1  | 12.8  | 13.7  | 15.1  | 16.6  | 17.7  | (33) |  |
|                                                                        |             | MCDB   | 25.7  | 25.2  | 23.3  | 21.2  | 18.2  | 15.7  | 15.0  | 16.0  | 17.7  | 20.3  | 23.0  | 24.3  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.3  | 10.8  | 10.4  | 10.1  | 8.8   | 8.0   | 7.8   | 8.7   | 9.8   | 10.6  | 11.1  | 11.0  | (35) |  |
|                                                                        |             | MCDBR  | 18.6  | 17.0  | 16.0  | 13.8  | 11.9  | 10.2  | 10.2  | 11.8  | 13.8  | 16.4  | 17.4  | 18.2  | (36) |  |
|                                                                        |             | MCWBR  | 6.3   | 5.7   | 5.2   | 5.4   | 5.4   | 5.5   | 5.8   | 6.6   | 6.7   | 7.1   | 6.6   | 6.7   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 14.7  | 13.7  | 12.1  | 10.9  | 9.3   | 8.5   | 8.4   | 10.5  | 11.6  | 13.5  | 14.3  | 15.0  | (38) |  |
|                                                                        |             | MCWBR  | 5.9   | 5.7   | 5.0   | 5.3   | 5.3   | 5.6   | 5.5   | 6.4   | 6.4   | 6.6   | 6.3   | 6.3   | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.354 | 0.348 | 0.329 | 0.320 | 0.301 | 0.292 | 0.293 | 0.302 | 0.315 | 0.332 | 0.337 | 0.336 | (40) |  |
|                                                                        | taud        |        | 2.520 | 2.551 | 2.592 | 2.609 | 2.633 | 2.628 | 2.622 | 2.585 | 2.542 | 2.511 | 2.522 | 2.548 | (41) |  |
|                                                                        | Ebn at Noon |        | 984   | 969   | 953   | 907   | 875   | 858   | 876   | 913   | 951   | 973   | 994   | 1005  | (42) |  |
|                                                                        | Edh at Noon |        | 112   | 105   | 94    | 82    | 71    | 67    | 70    | 82    | 97    | 108   | 111   | 109   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 7.35  | 6.41  | 5.28  | 3.97  | 2.94  | 2.52  | 2.73  | 3.62  | 4.86  | 6.02  | 6.92  | 7.58  | (44) |  |
| (45)                                                                   | RadStd      |        | 0.40  | 0.44  | 0.34  | 0.20  | 0.17  | 0.09  | 0.14  | 0.17  | 0.19  | 0.38  | 0.45  | 0.46  | (45) |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.27 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (6 neighbors) | +0.23 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -60     | +93     | +44     |

Nomenclature: See separate page